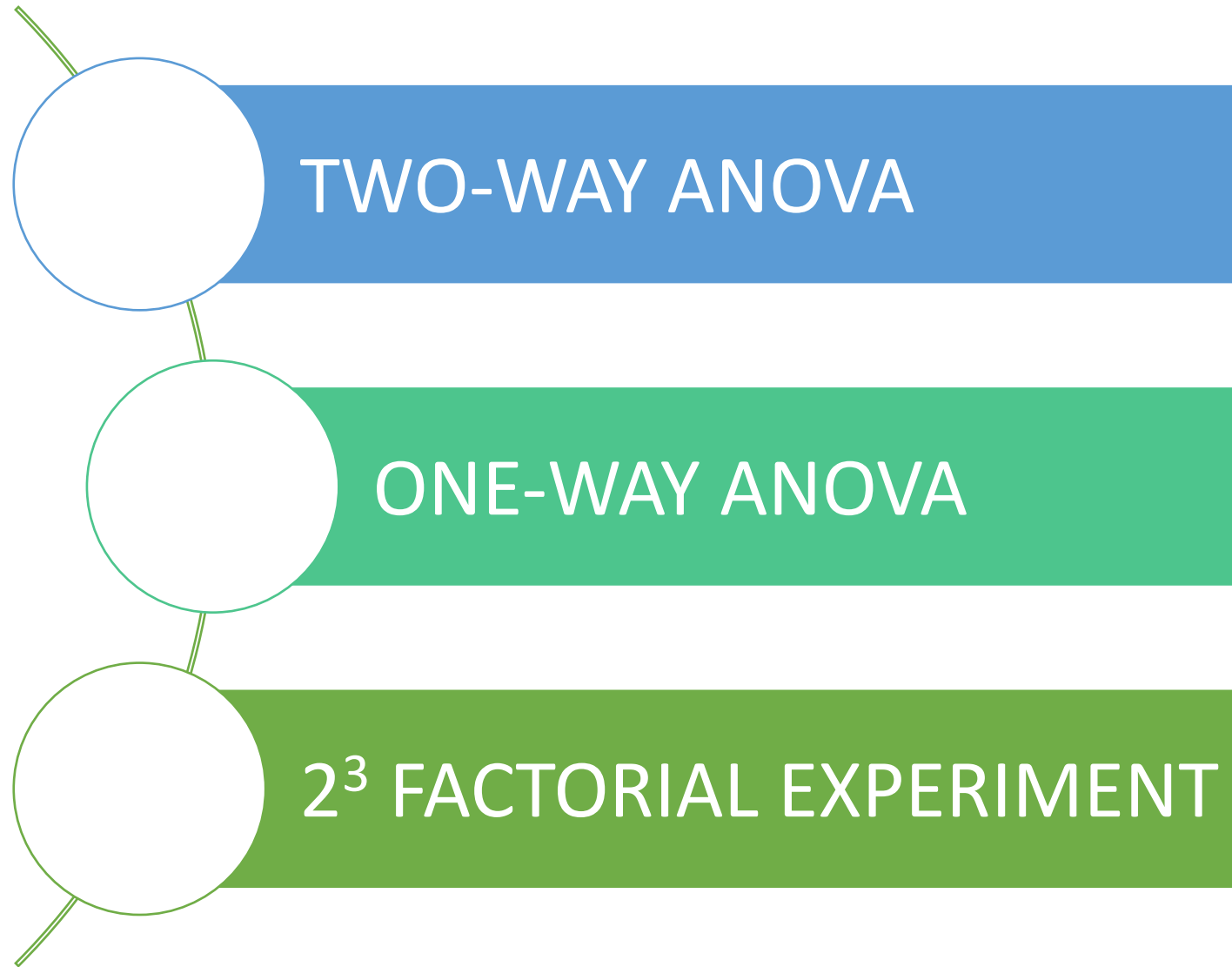


Stat 3113: Lesson 30

Examples!!!

PLAN



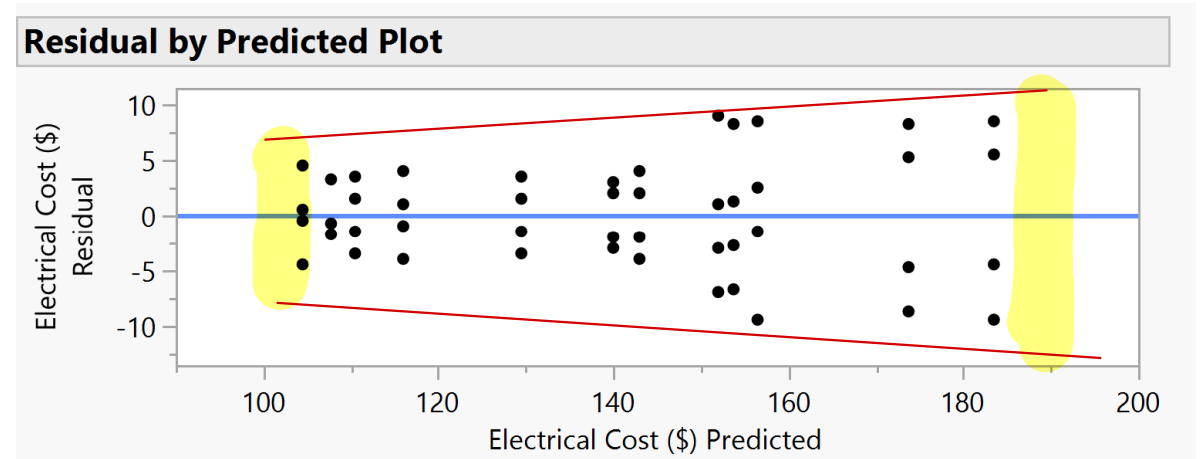
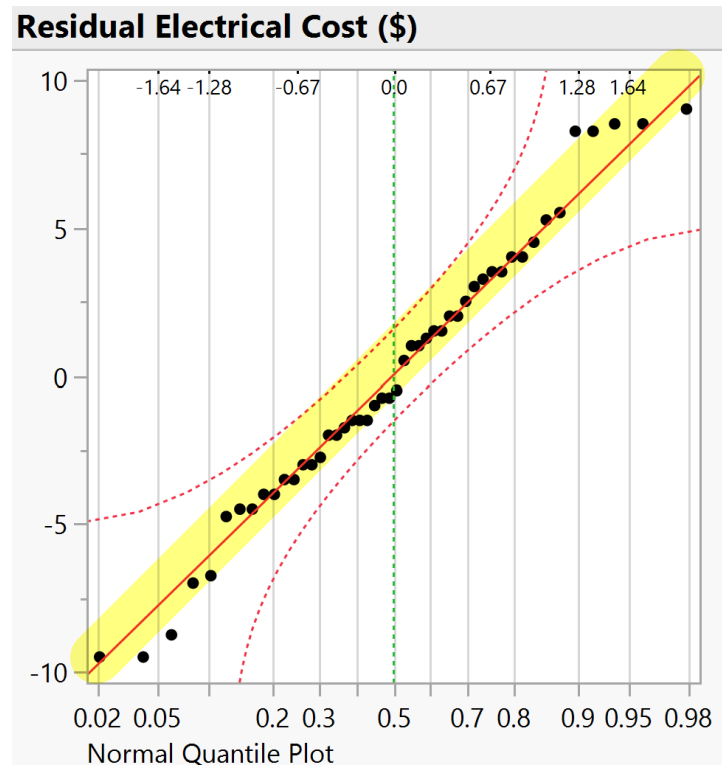
Example 1:

- In a particular city, a contractor desired to know if the **type of outer covering** on a house and the **type of air conditioning used** affect **electrical costs**. He randomly selected **48 homes** that fit into the levels of covering and type of air conditioning in which he was interested. He held variables such as house size and age constant. He measured the **total electrical cost per house for the month of July**. Data is on Canvas called *electric*.
- **Run an appropriate analysis. Use $\alpha = 0.05$ for all tests.**

Example 1:

- **Response variable:** electrical costs (\$)
- **Factors:** covering type, AC type
- **Levels:** covering – brick, wood, steel, vinyl
 - AC – heat pump, regular, none
- **Replicates:** 4 of each treatment

Check Conditions:



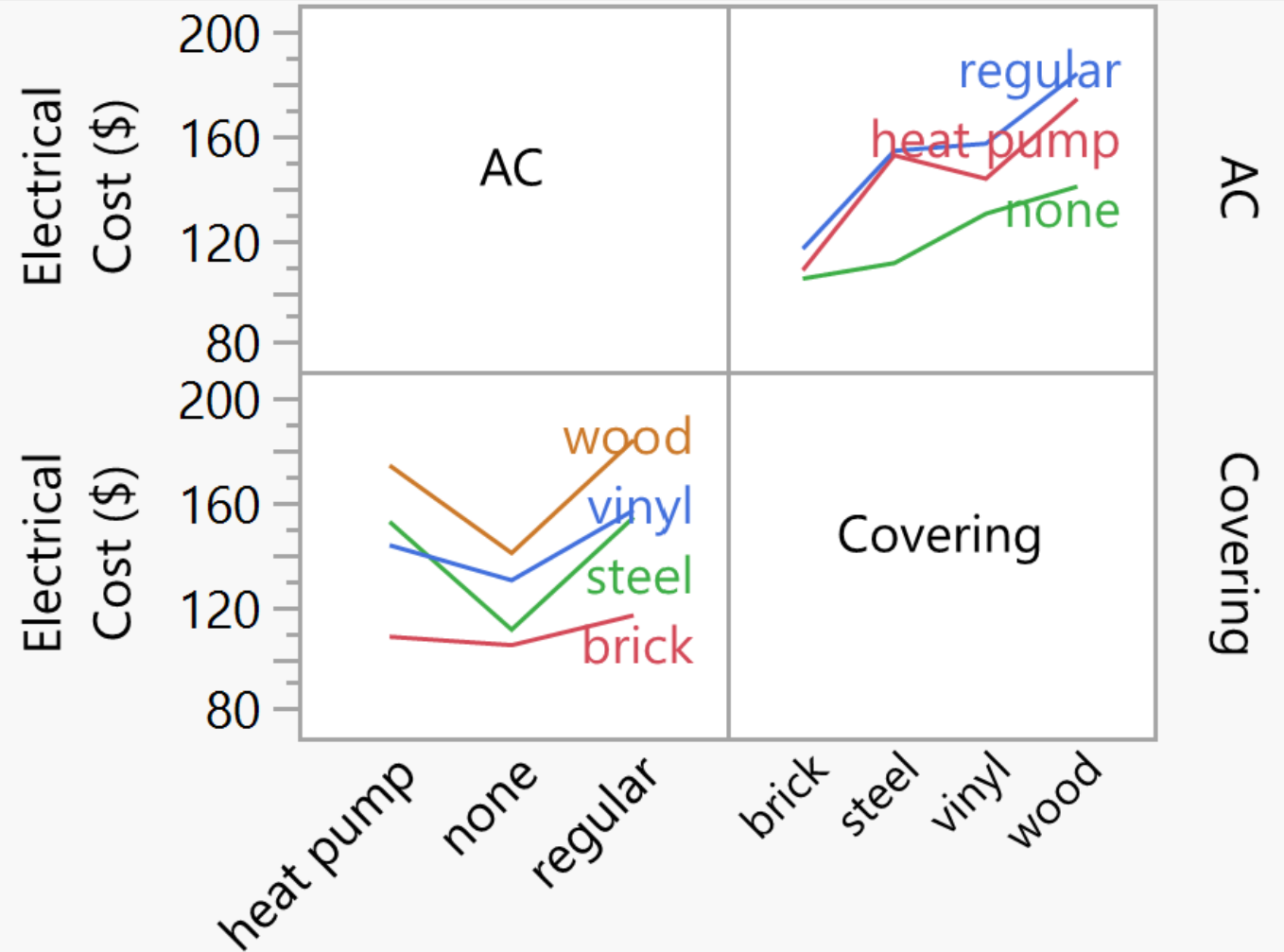
Analysis

Effect Tests

Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
AC	2	2	8419.042	143.6084	<.0001*
Covering	3	3	19279.063	219.2360	<.0001*
AC*Covering	6	6	2283.125	12.9815	<.0001*

Analysis

Interaction Profiles



Final Conclusions:

- There is a significant interaction between air conditioning type and covering type ($p\text{-value} < 0.0001$).
- The main effects for air conditioning type and covering type cannot be tested.
- Air conditioning type and covering type interact to significantly impact the average electric bill of homes.
- Constant variance of errors might be an issue.

Example 2:

- A pharmaceutical company conducts an experiment to test the effect of a new cholesterol medication. The company selects 15 subjects randomly from a larger population. Each subject is randomly assigned to a treatment group. Within each treatment group, subjects receive a different dose of the new medication. After 30 days, doctors measure the cholesterol level of each subject. Data is on Canvas called *cholesterol*.
- Run an appropriate analysis. Use $\alpha = 0.05$ for all tests.

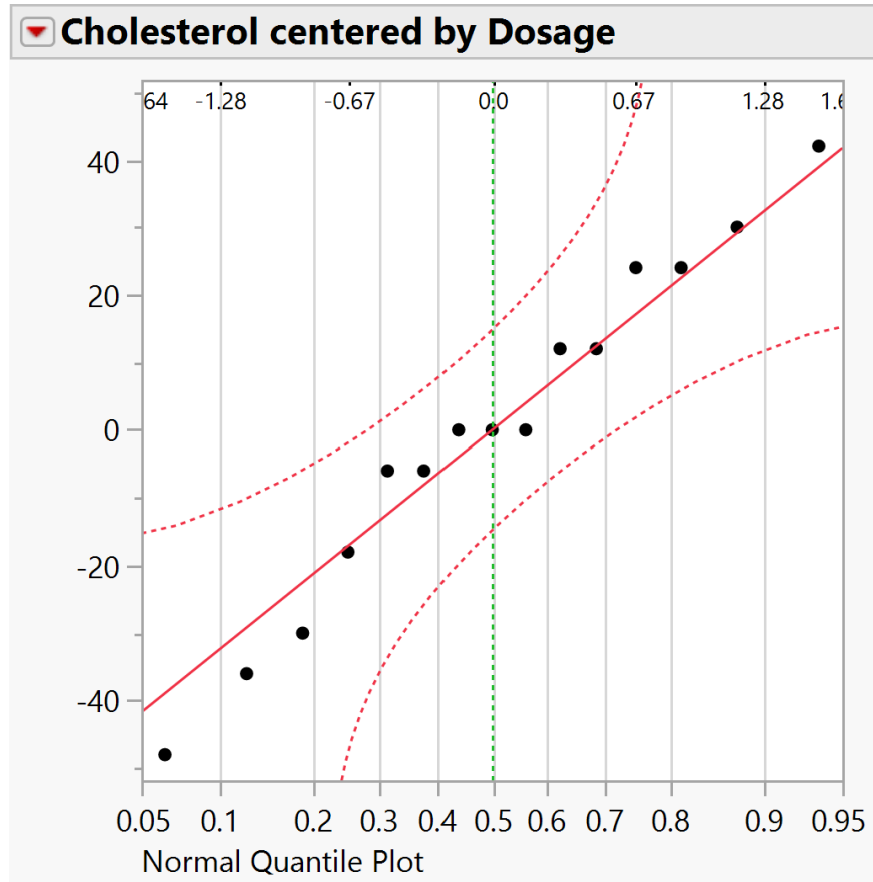
Response variable:
cholesterol level

Factor: medication amount

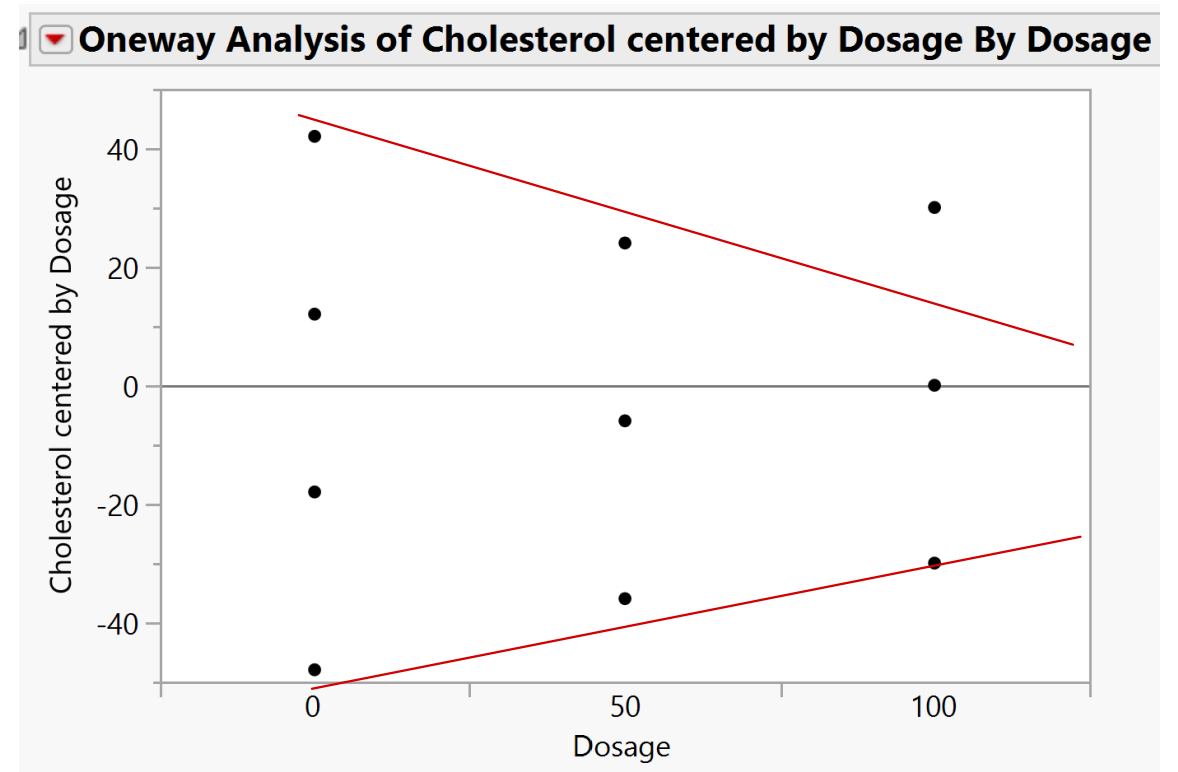
Levels: 0mg, 50mg, 100mg

Dosage		
Group 1, 0 mg	Group 2, 50 mg	Group 3, 100 mg
210	210	180
240	240	210
270	240	210
270	270	210
300	270	240

Check Conditions:



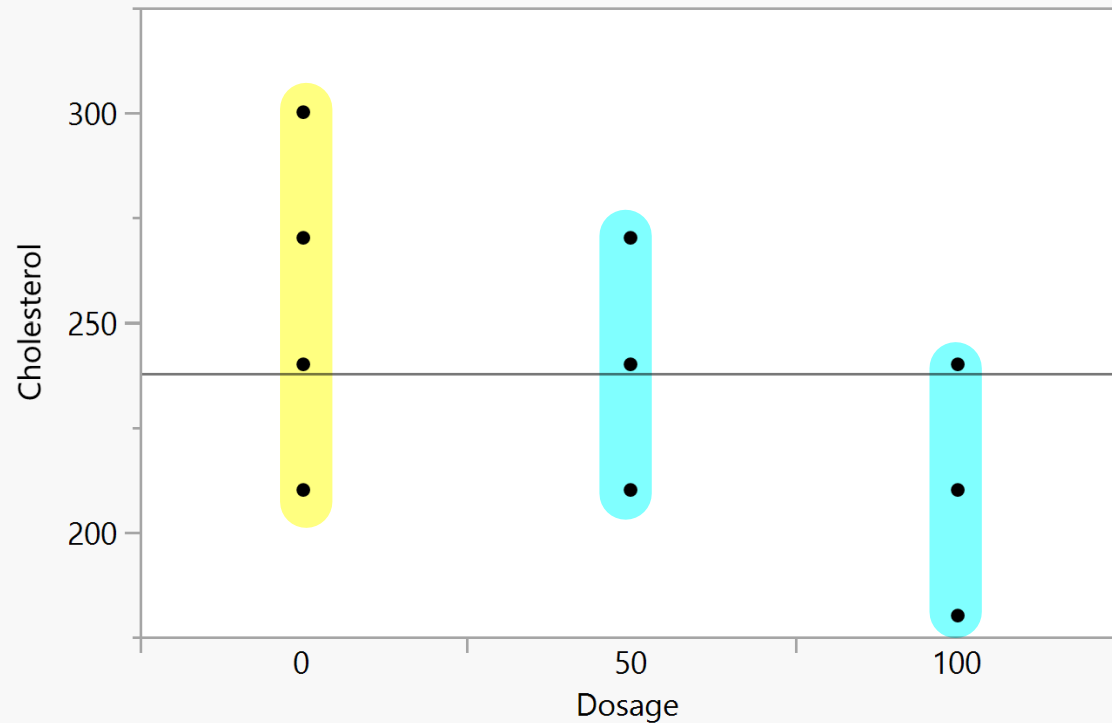
Normality holds ✓



Constant Variance seems to hold. ✓

Analysis

☒ Oneway Analysis of Cholesterol By Dosage



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio	Prob > F
Dosage	2	6240.000	3120.00	4.1600	0.0424*
Error	12	9000.000	750.00		
C. Total	14	15240.000			

Analysis

Connecting Letters Report

Level		Mean
0	A	258.00000
50	A B	246.00000
100	B	210.00000

Levels not connected by same letter are significantly different.

Ordered Differences Report

Level	- Level	Difference	Std Err Dif	Lower CL	Upper CL	p-Value
0	100	48.00000	17.32051	1.7931	94.20690	0.0417*
50	100	36.00000	17.32051	-10.2069	82.20690	0.1362
0	50	12.00000	17.32051	-34.2069	58.20690	0.7721

→ Reject H_0 !

Final Conclusions:

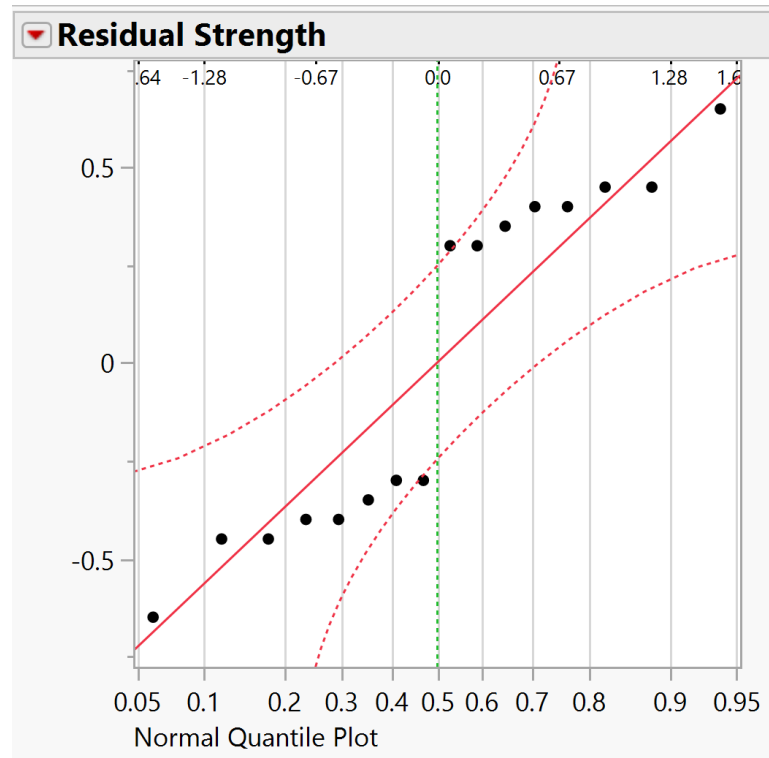
- At least one dosage resulted in a different mean cholesterol levels.
- There is a significant difference between mean cholesterol level of patients taking 0mg vs 100mg.
- Specifically, the mean cholesterol is lower for those taking 100mg versus 0mg.
- No other significant differences detected.

Example 3:

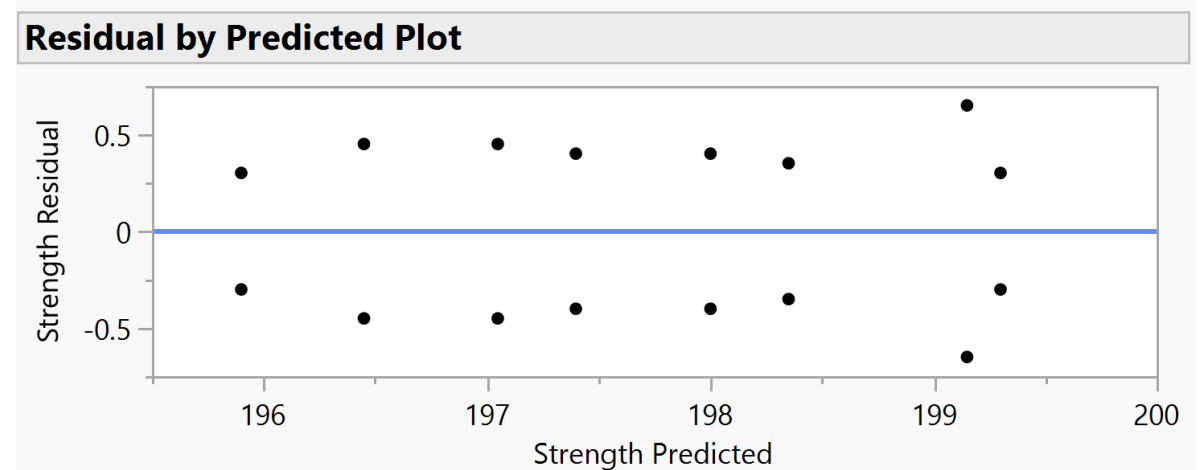
- A paper company is studying what impacts the strength of the paper they produce. They run an experiment varying percentage of hardwood concentration in raw wood pulp (4%, 8%),
- the vat pressure (500, 650), and the cooking time (3 hrs, 4 hrs). Data is on Canvas called *paper*.
- Run an appropriate analysis. Use $\alpha = 0.05$ for all tests.

-
- **Response variable:** paper strength
 - **Factor:** hardwood concentrations, vat pressure, cooking time
 - **Levels:** hardwood concentrations – 4%, 8%
 - vat pressure – 500, 650
 - cooking time – 3 hrs, 4 hrs
 - **Replicates:** 2

Check Conditions:



Normality seems to be violated!



Equal variance seems to hold!

Analysis

Effect Tests					
Source	Nparm	DF	Sum of Squares	F Ratio	Prob > F
Concentration	1	1	0.3600000	0.9931	0.3482
Pressure	1	1	2.4025000	6.6276	0.0329*
Concentration*Pressure	1	1	3.6450000	10.0552	0.0132*
Time	1	1	3.6100000	9.9586	0.0135*
Concentration*Time	1	1	1.2012500	3.3138	0.1062
Pressure*Time	1	1	0.1800000	0.4966	0.5010
Concentration*Pressure*Time	1	1	3.0625000	8.4483	0.0197*

→ STOP!

Final Conclusions:

- There is a significant interaction between hardwood concentration, vat pressure, and cooking time ($p\text{-value} < 0.0197$).
- No main effects can be tested
- Hardwood concentration, vat pressure, and cooking time interact to significantly impact the average paper strength.
- Normality is possibly problematic.



Reading

- CHAPTER 9 – SECTIONS 9.1, 9.2, 9.3, 9.5

Next Topic



Professor Koob's Projects Data Analysis



Chapter 9 – Sections 9.1, 9.2, 9.3, 9.5